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MFEFGAN: Multi-scale Feature Extraction and Fusion based Generative Adversarial Networks for Low-Light Enhancement of Remote Sensing Images

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***Abstract*—With the advancement of remote sensing technology, remote sensing images are gradually being widely used, low-light enhancement of remote sensing images is crucial for enhancing image quality and ensuring the accuracy of information. The accuracy of remote sensing image recovery is affected by the fact that existing algorithms lack sufficient semantic information and are not supervised by data pairs. Therefore, this article proposes a multiscale feature extraction and fusion network to enhance the semantic information contained in high-level features and use image pairs as supervised. First, the network downsamples the input image eight times to global features. Second, the feature fusion unit fuses the local features of different scales with the global features. Third, the network uses jump connections to send the fused features to the corresponding layer in the decoder. Finally, the PatchGAN discriminator is introduced to discriminate the generated image. It can directly learn the mapping relationship between input and output images without relying on complex radioactive transfer models, atmospheric parameters, surface reflection characteristics, etc. To validate the performance of this method, extensive comparisons are conducted with four traditional methods and three deep learning approaches. Experimental results show that the network significantly improves low-light remote sensing images, not only enhancing image contrast but also providing excellent color restoration and detail preservation.**

***Index Terms*—radiometric correction, remote sensing image, multi-scale, feature extraction**

I. Introduction

The rapid development of remote sensing technology allows us to obtain high-quality images of the earth's surface in an unprecedented way, providing rich information for environmental monitoring [1], [2], resource exploration [3], earth science [4], [5] and other fields. However, in the application of remote sensing images, the influence of radiation differences causes changes in contrast and brightness, which in turn leads to low-light phenomena in remote sensing images, seriously affecting the visibility and interpretability of the image [6]. Therefore, accurate low-light enhancement of remote sensing images has become a key step to ensure data quality and improve the accuracy of subsequent applications [7].

The goal of image enhancement (IE) is to improve the visual interpretation of images and provide clearer cues for further processing and analysis [8], [9]. Over time, numerous methods for enhancing low-light images have emerged, achieving great success in the fields of image processing and remote sensing. Among them, histogram equalization (HE) and its variants, such as contrast-limited adaptive histogram equalization (CLAHE) [10], are classic traditional techniques for enhancing image contrast. However, a significant drawback of HE is its tendency to distort colors [11] and excessively emphasize contrast [12]. Retinex theory has gained considerable attention as a biologically meaningful method for enhancing low-light images. The two most prevalent forms of this theory are SSR [13] and MSR [14]. However, these methods have been shown to have shortcomings in color reproduction. To address this issue, Multi-Scale Retinex with Color Restoration (MSRCR) [15] introduces a color restoration factor that adjusts color distortion caused by contrast enhancement in local image regions, improving color representation. Additionally, LIME employs

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conventional Lagrangian multipliers to enhance image details [16], [17]. However, the SSR model tends to produce halo-like artifacts in high-contrast areas [18], leading to the development of improved illumination estimation methods, such as Discrete Wavelet Transform and Singular Value Decomposition (DWT-SVD) [19], and other machine learning approaches.

In the area of low-light remote sensing image enhancement, most research has primarily concentrated on traditional and machine learning techniques, while deep learning methods have not been sufficiently explored. Convolutional Neural Networks (CNNs) are typically used to extract features from input images for processing. However, compared to CNNs, Generative Adversarial Networks (GANs) leverage a dynamic interplay between generators and discriminators through alternating training processes. This results in GAN-generated images of superior quality. Consequently, this paper proposes the application of advanced features within the GAN framework to enhance low-light remote sensing images. By utilizing global features to strengthen local ones at every level, richer semantic information is obtained, leading to significant improvements in tasks such as detection [17] and segmentation [18].

In this article, we propose a novel low-light enhancement method for remote sensing images, namely multi-scale feature extraction and fusion based generative adversarial network (MFEFGAN). MFEFGAN extracts information at more scales and integrates global features and local features at different scales to obtain richer semantic information. We conducted a series of experiments on the WHU-OPT-SAR dataset, comparing our design quantitatively and qualitatively with seven methods, and results show that our method achieves significant progress in low-light enhancement of remote sensing images.

II. Proposed Method

The design of this paper is based on the Conditional Generative Adversarial Network (CGAN), which consists of three modules: a generator, a discriminator, and a loss function. Among these, adversarial loss is more effective for image generation than Euclidean loss in DCNNs [20]. The input consists of a pair of images: a low-light image and its corresponding reference image. The generator, denoted as *G*, produces an image that resembles the reference image, attempting to deceive the discriminator *D*. The discriminator *D* aims to distinguish whether the image is real or fake. Through this adversarial process, *G* and *D* continuously update each other, resulting in progressively more realistic output images as *G* and *D* are trained iteratively. The overall network framework is shown in Fig. 1.

A. Generator

The generator is based on an encoder-decoder structure, while multi-scale feature extraction and feature fusion units are introduced and innovatively integrated [21], their network architecture is shown in Fig. 2. The effectiveness of these designs is demonstrated in the experimental section.

1) Global Features and Local Feature Extraction: Global features usually refer to the features extracted from the whole image, which represent the overall content and cover the whole image information. The opposite are local features, which focus on local areas in the image, such as corner points, edges, etc. Unlike ordinary average pooling, we perform layer-by-layer downsampling operations on the input features by convolution until the output has dimension $1\times1\times c_g$, where c_g represents the number of channels of the global features. Gradual downsampling has two advantages over simple average pooling: first, it can be more flexible to extract feature maps at different resolutions, and second, the number of feature maps at different resolutions can be freely chosen. Thus, this approach better preserves the edge detail information of the object and thus better captures different features in the image. Since the encoder part of the generator acts as a feature extractor, and to make the network deeper and easier to optimize, we use a residual building block similar to the one proposed in [22].

2) Upsampling of the Decoder: We employ a parameterized upsampling method through an inverse convolution operation. Each of its parameters is trainable, allowing the model to adapt to the data's features and learn the complex patterns present. This method can handle inputs and outputs of varying scales, providing flexibility in processing feature maps of different sizes. Additionally, the inverse convolution operation can be used interchangeably with the convolution operation, offering the neural network an improved structure for processing contextual information. This structure enables the network to learn higher-level features and achieve better accuracy.

3) Fusion of Global and Local Features: In this step, we propose a unit that dynamically fuses global and local features. When performing feature fusion at the ith scale, the dimensions of the local features are $h_i \times w_i \times c_i$, and the dimensions of the global features are $1 \times 1 \times c_g$. First, we extract the number of channels of the global features using a 1×1 convolutional layer that learns the weights. We dynamically extract the most useful information for the local features from this. The parameters of the convolutional layer are set as follows: The kernel size is 1×1, the stride is 1, the number of input channels is c_g, and the number of output channels is c_i. Denote f_{conv}, f_{copy}, $f_{reshape}$ and f_{concat} as convolution, replication, shaping, and concatenation operations, respectively. The feature fusion unit operates as described in f_1 through f_2. The output f_1 after convolution plus replication can be denoted as:

$$f_1 = f_{copy}(f_{conv}(F_g, w), num = h_i \times w_i) \tag{1}$$

where F_g and w represent the global features and learnable weights. The convolution is performed on F_g, where the convolution kernel size is 1×1, and then the convolved feature map is copied $h_i \times w_i$ times. The output f_2 after shaping plus concatenation can be denoted as:

$$f_2 = f_{concat}(f_{reshape}(f_1, size = (h_i, w_i, c_i)), F_l) \tag{2}$$

where F_l is the local feature map, f_1 is reshaped to (h_i, w_i, c_i) dimensions and spliced with F_l in the channel dimension. The

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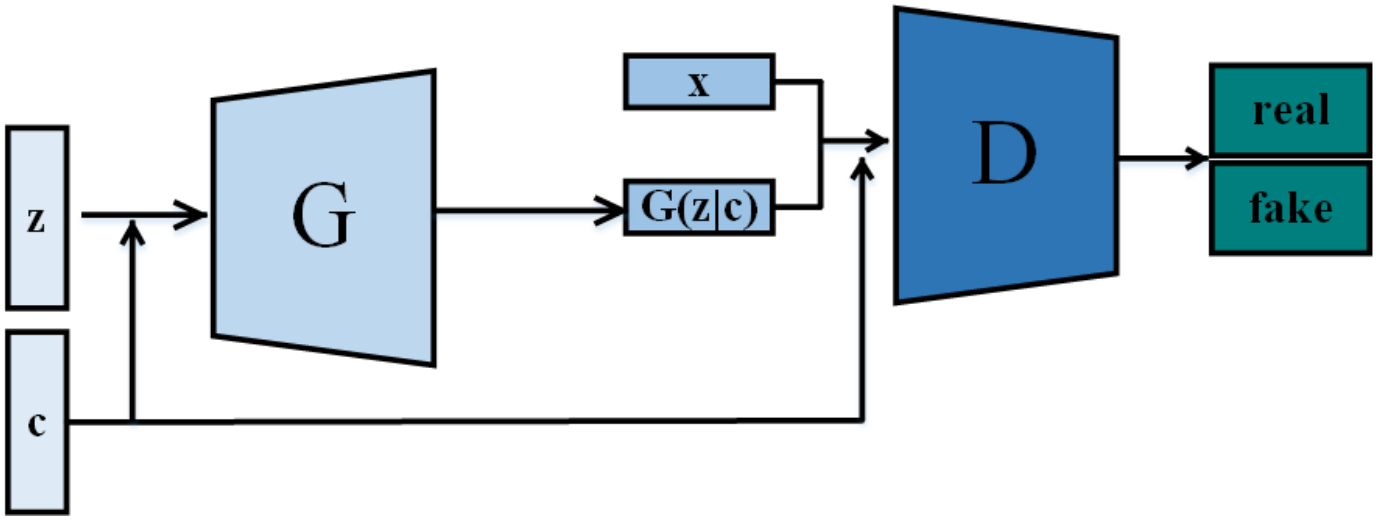


Fig. 1: Overall network framework diagram: Structure of a conditional generative adversarial network, where the generator *G* takes random noise z and condition c as input to produce samples $G(z \mid c)$,and the discriminator *D* determines whether the input is real or fake.

fused f_2 is fed to the corresponding layer in the decoder via a jump connection.

B. Discriminator

We used PatchGAN as a discriminator [20]. PatchGAN is designed to recognize whether each $N \times N$ block in an image is real or generated by a generator *G*. In this case, the input image is mapped into a 256 × 256 matrix after convolution with 3 discriminating blocks, PatchGAN tries to determine the authenticity of all blocks, the final decision is made by averaging the realism of all blocks. The generated image is only considered real if almost all image blocks have good, less blurred details. Since the network is trained in an adversarial manner, the generator is pushed to focus more on the high frequency part of the image (i.e., the image details), thus generating better image details to fool the discriminator. PatchGAN requires fewer convolutional layers compared to the overall image discriminator, its network architecture is shown in Fig. 3.

C. Objective Function

Vanilla GAN loss is one of the most fundamental loss functions in GANs [23]. It utilizes Binary Cross Entropy (BCE) as the loss function between the generator and the discriminator. The objective of the vanilla GAN loss is to make the samples generated by the generator appear indistinguishable from real samples in the eyes of the discriminator, while also enabling the discriminator to accurately distinguish between real and generated samples. For the sample *x* generated by the generator *G* and the real sample *y*, the loss function can be expressed as:

$$\begin{aligned} Loss_{GAN}(G,D) = {} & E_{y\sim pdata(y)}\left[\log D(y)\right] \\ & + E_{x\sim pgen(x)}\left[\log(1-D(G(x)))\right] \end{aligned} \tag{3}$$

During the training process, the generator aims to minimize *log(1 - D(G(x)))* as much as possible, making it difficult for the discriminator to distinguish between the generated samples and the real ones. On the other hand, the discriminator's goal is to maximize *log(D(y))* and *log(1 - D(G(x)))* in order to classify real and generated samples as accurately as possible.

Adversarial loss quantifies the Wasserstein distance, which measures the distributional difference between the generated image and the real image. In contrast, traditional metrics such as L_2 or L_1 loss measure pixel-level differences. The study in [24] highlighted the combination of adversarial loss with traditional metrics for image-to-image translation. Notably, the L_1 loss, as discussed in [20] and [24], tends to produce less ambiguous results compared to L_2 loss, which motivates its integration. The formula for L_1 loss is as follows:

$$\mathcal{L}_{L_1}(G) = E_{x,y}[\| \, y - G(x) \, \|_1] \tag{4}$$

The overall objective function loss $\mathcal{L}^*$ is (where λ is the weighting factor):

$$\mathcal{L}^* = \min_G \max_D Loss_{GAN}(G,D) + \lambda \mathcal{L}_{L_1}(G) \tag{5}$$

III. Experiments And Analysis

A. Experimental Setting

*1)*Dataset: Unlike other computer vision tasks, one of the challenges in remote sensing image processing is the limited availability of public datasets. To overcome this problem, this article is based on the WHU-OPT-SAR dataset mentioned in [25] and performs further processing to generate the required remote sensing image data. First, the RGB image is extracted from the optical image. Then, a program is used to randomly reduce the pixel value of the RGB image to simulate the low light phenomenon caused by the remote sensing image. Finally, the image is cropped to 600×600 and spliced to the

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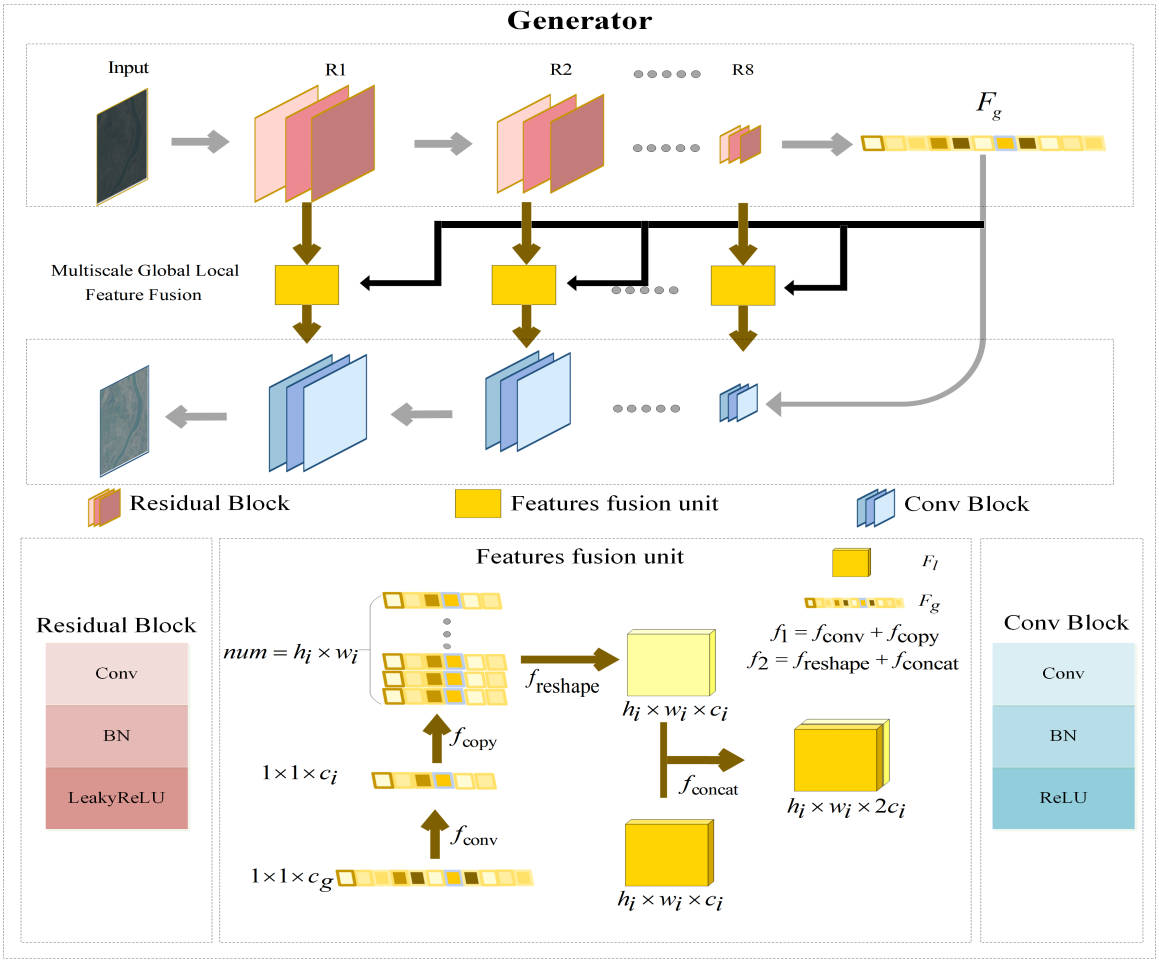


Fig. 2: Generator which consists of the encoder, decoder, and Feature fusion unit. The encoder realizes the extraction of multiple scale features. The decoder realizes the reconstruction of multi-scale features. The feature fusion unit realizes the fusion of multi-scale local features and global features.

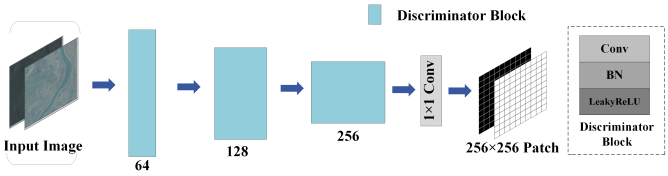


Fig. 3: PatchGAN based discriminator. PatchGAN discriminator enhances fine-grained image discrimination in GANs by evaluating the realism of local patches rather than the entire image. By focusing on local structures and textures, it encourages the generator to produce more detailed and realistic outputs, significantly improving the preservation of high-frequency details in complex image synthesis tasks.

original image, and from these images, we randomly select 4400 image pairs as the training set and the remaining 1100 image pairs as the validation and test sets.

*2)*Training Settings: In our experiments, we set $\lambda = 10$. These values are chosen based on basic hyperparameter tuning. We applied the Adam solver with a learning rate $lr = 0.0002$, $\beta_1 = 0.5$, and $\beta_2 = 0.999$. The batch size was 1. The network was trained for 100 epochs.

*3)*Comparative Methods: To verify the validity of this study, we compared the proposed method with recent conventional methods as well as the latest research methods published in top conferences or journals. Specifically, we compared seven methods, including different types of methods such as histogram type CLAHE [10], retinex type SSR [13] and MSRCR [15], illumination map estimation and retinex type LIME [16], as well as the deep learning methods of RSCNN [26], Cyclegan [27] and Pix2Pix [28]. These models used the default settings recommended by the author in the experiment.

*4)*Evaluation Metrics: We test the trained models on a test dataset and make a quantitative comparison, we use three commonly used image quality evaluation metrics, Peak Signal Noise Ratio (PSNR), Structural Similarity Index (SSIM) and Mean Square Error (MSE).

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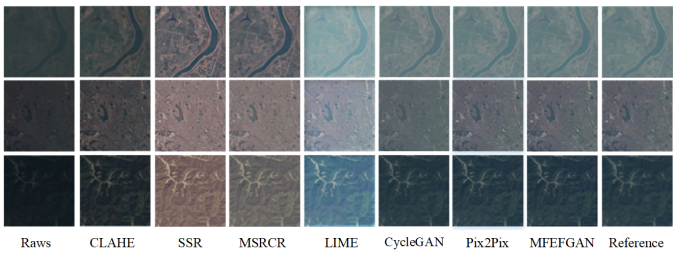


Fig. 4: Visualization results of image enhancement using different methods, including CLAHE [10], SSR [13], MSRCR [15], LIME [16], CycleGAN [27], and Pix2Pix [28].

TABLE I: The average SSIM and PSNR of the test results.

Methods	SSIM	PSNR
CLAHE [10]	0.833	18.880
SSR [13]	0.750	18.084
MSRCR [15]	0.782	27.898
LIME [16]	0.706	27.433
RSCNN [26]	0.825	28.194
CycleGAN [27]	0.852	30.375
Pix2Pix [28]	0.972	36.724
MFEFGAN(ours)	**0.981**	**37.436**

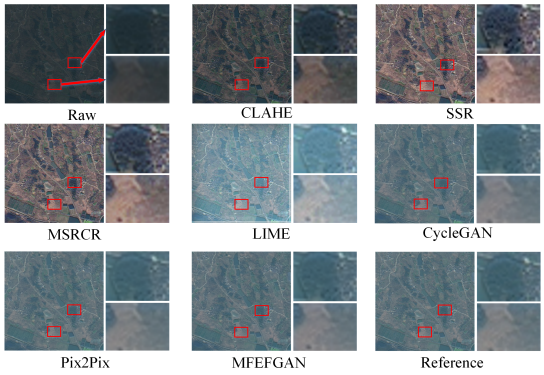


Fig. 5: A zoomed-in view comparison of different methods, including CLAHE [10], SSR [13], MSRCR [15], LIME [16], CyclGAN [27] and Pix2Pix [28].

TABLE II: Comparison and evaluation of different generator structures using this method.

Evaluations	Res-Net [22]	U-Net [28]	Ours-5	Ours-8
SSIM	0.87	0.97	0.93	**0.98**
PSNR	27.92	36.72	30.11	**37.43**
MSE	0.00326	0.00022	0.00196	**0.00018**

B. Results

Table I presents a quantitative comparison of MFEFGAN with seven other methods. The table displays the average values of SSIM and PSNR, indicating that deep learning methods outperform traditional methods. MFEFGAN performs exceptionally well in deep learning, achieving the highest SSIM and PSNR values, which are significantly better than the other methods. Fig.4 illustrates that MFEFGAN achieves the best image enhancement effect, which is close to the reference image. In the enlarged view of Fig.5, it is evident that the image processed by CLAHE is darker than the reference image. The processed images from SSR, MSRCR, and LIME show severe color distortion and introduce white noise dots. The processed image from SSR has a yellowish tone, while the processed image from LIME is brighter than the original reference image, giving the impression of being covered by fog. The images processed by CycleGAN, Pix2Pix, and MFEFGAN are color consistent with the reference image. However, in the quantitative analysis, Pix2Pix performs slightly worse than MFEFGAN. CycleGAN produces blurry results when trying to recover dark green scenes. In a word, our method achieves effective color correction results and smooth details without introducing unnecessary noise, unlike other methods.

C. Generator Structure Comparisons

To evaluate the effectiveness of our proposed generator structure, we compared it with other generator models, including resnet_9blocks [22], U-net [28], Ours-5(containing 5 scales of features), and Ours-8(containing 8 scales of features). The generator was the only component modified during training, with a batch size of 1 for 100 epochs. Table II presents the test results, which demonstrate that our proposed structure can achieve the highest PSNR with SSIM and the smallest MSE. Compared with Ours-5, Ours-8 achieves 5% higher SSIM, 7dB higher PSNR, and lower MSE, it demonstrates the importance of multiple scales. Overall, our generator outperforms the current mainstream Res-Net and U-Net generators in SSIM, PSNR, and MSE, which demonstrates the effectiveness of the proposed generator structure in terms of its learning ability and performance.

IV. Conclusion

This article proposes an end-to-end multi-scale feature extraction and fusion generative adversarial network model for obtaining corrected standard remote sensing images from low-light remote sensing images. The model uses a generative adversarial network structure to achieve end-to-end low-light enhancement of remote sensing images. Multi-scale feature extraction allows the network to capture broader and more global information. High scales are used to capture the overall structure and context, while low scales focus on details and textures. In addition, multi-scale feature fusion improves the quality and realism of the generated images. Compared with other traditional and deep learning methods, our resulting images are more natural, with realistic textures and vivid details. Our approach aims to reveal as many edge and structural features as possible, which is beneficial for subsequent high-level remote sensing image information retrieval tasks.

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